

BCEM Imp Topics

BT-204: Basic Civil Engineering & Engineering Mechanics

UNIT-WISE 5 MOST IMPORTANT TOPICS (PYQs 2019–2024 BASED)

UNIT–I: Building Materials & Construction

1. **Properties of Cement (physical + chemical)**

Asked in: **May 2019, Jun 2020, Nov 2022, Dec 2023**

2. **Workability of Concrete & Factors Affecting It**

Asked in: **Jun 2022, Nov 2019, Dec 2024**

3. **Nominal Mix of Concrete + Preparation, Compaction & Curing**

Asked in: **Jun 2022, Dec 2023, Dec 2024**

4. **Types of Foundations (Shallow & Deep)**

Asked in: **Nov 2019, Jun 2023, Dec 2024**

5. **Plastering & Pointing / Brick Masonry / Types of Stairs**

Asked in: **Jun 2020, Jun 2023, Nov 2022**



Pattern: Pure theory + short notes

UNIT–II: Surveying & Positioning

1. **Surveying – Definition, Principles & Classification**

Asked in: **May 2019, Jun 2020, Nov 2019**

2. **EDM (Electronic Distance Measurement)**

Asked in: **May 2019, Jun 2023, Dec 2023**

3. Levelling & Reciprocal Levelling (procedure + numericals)

Asked in: **Jun 2020, Jun 2022, Nov 2022, Dec 2024**

4. Survey Instruments (Level, Theodolite – parts & uses)

Asked in: **Nov 2019, Dec 2023, Nov 2022**

5. Survey Stations & Types of Surveying

Asked in: **Jun 2023, Nov 2019**



Pattern: Theory + 1 numerical almost every year

UNIT–III: Mapping & Remote Sensing

1. Contour Lines – Definition, Properties & Uses

Asked in: **Jun 2020, Jun 2023, Nov 2019, Dec 2023**

2. Contour Interval & Horizontal Interval

Asked in: **Nov 2022, Dec 2023**

3. Remote Sensing – Principles & Applications

Asked in: **May 2019, Jun 2020, Nov 2022, Dec 2024**

4. Measurement of Areas (Mid-ordinate, Trapezoidal, Simpson's Rule)

Asked in: **Jun 2022, Dec 2024**

5. Profile & Cross-Section Levelling

Asked in: **Jun 2022, Nov 2022**

UNIT–IV: Engineering Mechanics – Forces & Equilibrium

1. Coplanar & Concurrent Forces + FBD

Asked in: **May 2019, Jun 2020, Nov 2019, Jun 2023**

2. Lami's Theorem (statement + numerical)

Asked in: **Jun 2020, Nov 2022, Dec 2023**

3. Varignon's Theorem

Asked in: **Jun 2020**

4. Plane Truss – Method of Joints

Asked in: **May 2019, Jun 2022, Dec 2024**

5. Method of Sections (theory / numerical)

Asked in: **Jun 2023, Nov 2022**

UNIT–V: CG, MI, SFD & BMD

1. Shear Force Diagram (SFD) & Bending Moment Diagram (BMD)

Asked in: **EVERY YEAR (2019–2024)**

2. Support Reactions (Simply supported & Cantilever)

Asked in: **Nov 2022, Jun 2023**

3. Moment of Inertia of Plane Lamina (Rectangular, Composite)

Asked in: **May 2019, Dec 2023, Nov 2022**

4. Parallel Axis & Principle Axis Theorem

Asked in: **Jun 2020, Dec 2023**

5. Centroid, Centre of Gravity & Radius of Gyration

Asked in: **Nov 2019, Jun 2022, Jun 2023**
